

Chapter 8

1. closure:

For $\forall (g_1, g_2, \dots, g_n)$ and $(h_1, h_2, \dots, h_n) \in G \oplus H$,
their product $(g_1 h_1, g_2 h_2, \dots, g_n h_n) \in G \oplus H$ since
 G and H are closed groups

Associativity:

Since group operation is component wise, the group
is associative

identity:

Identity element in $\mathbb{Z}DP \alpha$ is just tuple of
identity element, where e_i represent identity of
 G_i .

$$(g_1, g_2, \dots, g_n) \cdot (e_1, e_2, \dots, e_n) =$$

$$(e_1, e_2, \dots, e_n) \cdot (g_1, g_2, \dots, g_n) = (g_1, g_2, \dots, g_n)$$

inverses:

For $\forall (g_1, g_2, \dots, g_n) \in G$, $\exists (g_1^{-1}, g_2^{-1}, \dots, g_n^{-1})$
 $\in G$, where g_i^{-1} is the inverse of g_i in G_i ; given
 G_i is a group

$$(g_1, g_2, \dots, g_n) \cdot (g_1^{-1}, g_2^{-1}, \dots, g_n^{-1}) =$$

$$(g_1^{-1}, g_2^{-1}, \dots, g_n^{-1}) \cdot (g_1, g_2, \dots, g_n) = (e_1, e_2, \dots, e_n)$$

3. Define mapping: $\phi: G \rightarrow G \oplus \{e_n\}$
by $\phi(g) = (g, e_n) \quad \forall g \in G$

O.P.:

$$\phi(gh) = (gh, e_n) = (g, e_n) \cdot (h, e_n) = \phi(g) \phi(h)$$

Thus ϕ is operation preserving

1-1:

$$\text{Let } \phi(g) = \phi(h). \text{ This means } (g, e_n) = (h, e_n)$$

Thus $g = h$. ϕ is injective

onto:

For $\forall (g, e_n) \in G \oplus \{e_n\}$, $\exists g \in G$ such that $\phi(g) = (g, e_n)$. This ϕ is surjective

$\therefore \phi$ is an isomorphism

4. Define mapping: $\psi: H \rightarrow \{e_n\} \oplus H$
by $\psi(h) = (e_n, h) \quad \forall h \in H$

O.P.:

$$\psi(hk) = (e_n, hk) = (e_n, h) (e_n, k) = \psi(h) \psi(k)$$

Thus ψ is operation preserving.

1-1:

$$\text{Let } \psi(h) = \psi(k). \text{ This means } (e_n, h) = (e_n, k)$$

Thus $h = k$. ψ is injective.

onto:

For $\forall (e_n, h) \in \{e_n\} \oplus H$, $\exists h \in H$ such that $\psi(h) = (e_n, h)$. Thus ψ is surjective.

$\therefore \psi$ is also an isomorphism

5. Assume for contradiction $\exists (a, b) \in \mathbb{Z} \oplus \mathbb{Z}$ such that $\mathbb{Z} \oplus \mathbb{Z} = \langle (a, b) \rangle$

Consider generators from both groups $(-1, -1), (1, -1), (-1, 1), (1, 1)$

All of these tuples fail to generate all element in the I.D.P set
 $\therefore \mathbb{Z} \oplus \mathbb{Z}$ is not cyclic

6. Show $\mathbb{Z}_4 \oplus \mathbb{Z}_2 \neq \mathbb{Z}_8 \oplus \mathbb{Z}_1$

Consider elements of order 4 in both groups

$$4 = \text{lcm}(4, 1)$$

$$= \text{lcm}(4, 2)$$

Count elements:

$$2 + 2 = 4$$

$$4 = \text{lcm}(4, 4)$$

$$= \text{lcm}(2, 4)$$

$$= \text{lcm}(4, 2)$$

$$= \text{lcm}(4, 1)$$

$$= \text{lcm}(1, 4)$$

Count elements:

$$(2 \cdot 2) + (1 \cdot 2) + (2 \cdot 1) + (2 \cdot 1)$$

$$+ (1 \cdot 2) = 12 \neq 4$$

$$\therefore \mathbb{Z}_4 \oplus \mathbb{Z}_2 \neq \mathbb{Z}_8 \oplus \mathbb{Z}_1$$

8. No. $\text{gcd}(3, 9) \neq 1$

9. $\mathbb{Z}_6 \oplus \mathbb{Z}_2$

$$|\langle (2, 1) \rangle| = 3 \cdot 2 = 6$$

$$|\langle (1, 0) \rangle| = 6 \cdot 1 = 6$$

14. D_n for $n \geq 3$ is a non-Abelian group

The rotation subgroup is Abelian while the other subgroup of order 2 is also Abelian

$\therefore D_n$ cannot be isomorphic to I.D.P of such groups

18. $\mathbb{Z}_{10} \oplus \mathbb{Z}_{10}$

$$|\langle (10, 10) \rangle| = 4 \cdot 3 = 12$$

$$\text{Let } H = \langle (20, 0), (0, 5) \rangle$$

$$H \cong \mathbb{Z}_2 \oplus \mathbb{Z}_6 \text{ with order } 12$$

Since $g \notin \langle (2, 6) \rangle \neq 1$, it is not a cyclic subgroup

20 Let $K = \langle (3, 0), (0, 2) \rangle$

$$|(3, 0)| = \text{lcm}(4, 1) = 4$$

$$|(0, 2)| = \text{lcm}(1, 9) = 9$$

$$|K| = 4 \cdot 9 = 36$$

$$K \cong \mathbb{Z}_4 \oplus \mathbb{Z}_9$$

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order	$\mathbb{Z}_{20}$	$\mathbb{Z}_{20}$
1	1	1
3	2	2
5	4	4
15	8	0

$$15 = \text{LCM}(15, 1) = 8 \cdot 1$$

$$= \text{LCM}(3, 5) = 2 \cdot 4$$

$$= \text{LCM}(15, 5) = 8 \cdot 4$$

$$\text{Count elements: } 8 + 8 + 32 = 48$$

b. # of distinct subgroups = $48 \div \phi(48) = 48 \div 8 = 6$

33 Let $H = \langle (400, 0), (0, 50) \rangle$

$$|(400, 0)| = \text{LCM}(2, 1) = 2$$

$$|(0, 50)| = \text{LCM}(1, 4) = 4$$

$$|H| = 8$$

$$H \cong \mathbb{Z}_2 \oplus \mathbb{Z}_4$$

37 Let $G = \{3^m 6^n \mid m, n \in \mathbb{Z}\}$ [show $G \cong \mathbb{Z} \oplus \mathbb{Z}$]

An element in G is of the form $3^m \cdot 6^n$ where $m, n \in \mathbb{Z}$
 $= 3^m \cdot 2^n \cdot 3^n = 2^n \cdot 3^{m+n}$

Define Mapping: $\phi: G \rightarrow \mathbb{Z} \oplus \mathbb{Z}$ by $\phi(3^m 6^n) = (n, m+n)$

D.P.:

Let $g_1 = 3^{m_1} \cdot 6^{n_1}$ and $g_2 = 3^{m_2} \cdot 6^{n_2}$

$$\begin{aligned} g_1 \cdot g_2 &= 3^{m_1+m_2} \cdot 6^{n_1+n_2} \\ &= 2^{n_1+n_2} \cdot 3^{m_1+m_2+n_1+n_2} \end{aligned}$$

$$\phi(g_1 g_2) = (n_1 + n_2, m_1 + m_2 + n_1 + n_2)$$

$$g_1 = 3^{m_1} \cdot 2^{n_1} \cdot 3^{n_1}$$

$$\phi(g_1) = (n_1, m_1 + n_1)$$

$$g_2 = 3^{m_2} \cdot 2^{n_2} \cdot 3^{n_2}$$

$$\phi(g_2) = (n_2, m_2 + n_2)$$

$$\phi(g_1) + \phi(g_2) = (n_1 + n_2, m_1 + n_1 + m_2 + n_2) = \phi(g_1 g_2)$$

ϕ is operation preserving

1-1:

$$\begin{aligned} \text{Let } \phi(g_1) &= \phi(g_2) \cdot \text{This means } (n_1, m_1 + n_1) \\ &= (n_2, m_2 + n_2) \end{aligned}$$

Thus $n_1 = n_2$.

Given $n_1 = n_2$, $m_1 = m_2$

So $g_1 = g_2$

ϕ is injective

onto

Let $(k, l) \in \mathbb{Z} \oplus \mathbb{Z}$ [find $g \in G$ such that $\phi(g) = (k, l)$]

Find m and n such that:

$$n = k$$

$$n + m = l$$

$$m = l - n = l - k$$

$$\begin{aligned} g &= 3^m \cdot 6^n = 3^{l-k} \cdot 6^k \\ &= 3^{l-k} \cdot 3^k \cdot 2^k \end{aligned}$$

$$\phi(g) = (k, l - k + k) = (k, l)$$

ϕ is surjective.

$$\therefore G \cong \mathbb{Z} \oplus \mathbb{Z}$$

b. No. Group G is cyclic group generated by element 3.

$$47. \text{Aut}(U(25))$$

$$\phi(25) = 20$$

$$U(25) \cong \mathbb{Z}_{20}$$

$$\text{Aut}(\mathbb{Z}_{20}) \cong U(20) \cong U(5) \oplus U(4)$$

$$\cong \mathbb{Z}_{5-1} \oplus \mathbb{Z}_2$$

$$= \mathbb{Z}_4 \oplus \mathbb{Z}_2$$

* 59.

* 61.

$$65. \text{Aut}(\mathbb{Z}_{105}) \cong U(105) = U(3 \cdot 5 \cdot 7) \cong \mathbb{Z}_2 \oplus \mathbb{Z}_4 \oplus \mathbb{Z}_6$$

$$6 = \text{LCM}(1, 1, 6) = 1 \cdot 1 \cdot 2$$

$$= \text{LCM}(2, 1, 6) = 1 \cdot 1 \cdot 2$$

$$= \text{LCM}(1, 2, 6) = 1 \cdot 1 \cdot 2$$

$$= \text{LCM}(1, 2, 3) = 1 \cdot 1 \cdot 2$$

$$= \text{LCM}(2, 2, 6) = 1 \cdot 1 \cdot 2$$

$$= \text{LCM}(2, 2, 3) = 1 \cdot 1 \cdot 2$$

$$= \text{LCM}(2, 1, 3) = 1 \cdot 1 \cdot 2$$

$$\text{Count elements} : 2 \cdot 7 = 14$$

* 71.

$$1000 / 4 = 125$$

$$U_{125}(1000) = \{ \underline{1}, \underline{251}, \underline{501}, \underline{751} \}$$